

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • GPA: 3.75/4.0

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

EXPERIENCE

TiMoto AI | Software Engineer

Sep 2025 – Present

- Built LLM-to-API integration layer for a 3-person team – **designed, deployed, and operated** the backend bridge between large language models and internal data ecosystems, achieving **100% eval success rate at sub-50ms p99**; resolved production deadlock via gRPC call-sequencing redesign
- Architected **vLLM inference engine with PagedAttention** for LLM serving – eliminated KV cache memory fragmentation under concurrent load; deployed with continuous batching achieving **zero OOM failures** at production traffic; paired with **LangChain-based RAG evaluation pipeline** for ground-truth grounding
- Built LLM-as-a-judge evaluation harness to measure generation quality and surface hallucination patterns – used to tune prompts, select models, and validate **production accuracy SLOs**
- Architected multi-AZ ECS Fargate with **Terraform IaC**, CloudWatch observability, and circuit breaker pattern – **99.9% uptime, 44% cost reduction** (\$40–60/month); auto-rollback on health check failure

Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers – shipped to Chrome stable channel serving **3B+ active users** at **sub-50ms p99, 10K+ req/sec**; selected Protobuf over custom serialization for schema evolution and cross-language compatibility
- Identified settings navigation as p99 bottleneck at 1,200ms; designed and implemented **lock-free concurrent trie search** eliminating mutex contention – **96% latency reduction**, zero production regressions
- Architected event-driven TypeScript/React system with observer pattern – accelerated feature delivery **68%** across **25K+ lines** of Chromium at **95% test coverage**; design documents reviewed by senior Chrome engineers

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on AWS (Python) – JWT over session auth for horizontal scalability; auto-scaling policy supporting **500+ concurrent users, 90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned with PostgreSQL indexing achieving **sub-100ms** for 10,000+ records

PROJECTS

Pulumi – Contributor | Go, TypeScript, IaC, GCP/AWS/Azure multi-cloud

github.com/pulumi

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning – under active review by core maintainers; studied Raft/Paxos consensus in distributed state synchronization layer
- Analyzed linearizability guarantees and correctness under concurrent operations – translates directly to orchestration framework reliability requirements

SKILLS

ML & AI Infrastructure: vLLM, PagedAttention, RAG Pipelines, LangChain, LangGraph, Function Calling, LLM-as-a-judge, LoRA/QLoRA, PyTorch

Cloud & Infrastructure: GCP (Vertex AI), AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, K8s, Docker, CI/CD, CloudWatch, Prometheus

Distributed Systems: gRPC, Protocol Buffers, Circuit breakers, Exactly-once semantics, Fault tolerance, Kafka, Multi-AZ, Event-driven architecture

Languages: Python, Java, C++, Go, TypeScript, JavaScript, SQL, Bash, Rust

Frameworks & Databases: Django, FastAPI, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor